The Burden of Psychological Pain in Laying Hens: Behavioral Deprivation

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Over millions of years, evolution has shaped the motivation of animals to explore the environment for critical resources and information, to be alert for subtle signs of danger and to opportunities, and to constantly resolve fluid social standings. Hence, it comes at no surprise that severe limitations on the realization of these motivations imposed on confined animals can be major sources of poor welfare. This is of special relevance for animals kept in the barren and tight spaces that are typical of intensive production systems, as is the case of caged housing. Despite over 8,000 years of domestication [1], there is no evidence that the cognitive abilities and complex behavior of farmed breeds has changed substantially compared to that of their wild ancestors [2]. Given the opportunity, modern laying hens will forage and explore the environment for several hours per day [3], go through the typical behavioral sequence involved in the search for and the construction of a nest, roost at night, dustbathe, flap their wings, and move around - in the same way that their ancestor, the red jungle fowl, do. However, under intensive housing conditions, the motivation to express these behaviors is neither properly fulfilled, nor are hens able to exert any control over the circumstances affecting their well-being.

The International Association for the Study of Pain (IASP) defines pain as "*an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage*" [4]. Accordingly, any unpleasant experience that potentially threatens the integrity of an individual, or her kin, immediately or in the future, can be treated as a painful experience. Similarly, experiences of psychological pain (also referred to as

(If you prefer to have this material in e-book format or printed format, please find it at https://www.amazon.com/Quantifying-Pain-Laying-Hens-comparative-ebook/dp/B09BR69NV6/)

Please cite it as
Schuck-Paim C., Alonso WJ 2021. The Burden of Psychological Pain in Laying Hens: Behavioral Deprivation . In: Schuck-Paim C, Alonso WJ, editors. Quantifying Pain in Laying Hens. Independently published. https://tinyurl.com/bookhens

emotional pain), namely negative affective states such as fear, anxiety, boredom, helplessness and depression, can be operationally classified using the same criteria (Chapter 1 [5]).

Here we apply the analytical framework described in Chapter 1 [5] to the measurement of pain of the latter nature (psychological), particularly that associated with the deprivation of various behavioral needs as experienced by commercial laying hens. To this end, we examine the possible intensity of the pain suffered as a consequence of each type of behavioral deprivation, in addition to its duration and temporal evolution. Although the deprivation of highly motivated behaviors can affect welfare in multiple ways (e.g., leading to the development of abnormal behaviors such as injurious pecking - Chapter 4 [6] - or increasing the likelihood of disease through immunosuppression [7]), our interest here is only in the potential psychological pain caused in each case.

Measuring the burden of psychological pain in laying hens

Despite a long history of domestication and intense artificial selection for elevated egg production under confined conditions, laying hens still retain a behavioral repertoire comparable to that of their wild ancestors. Although not every behavior performed in the wild is necessarily important for the welfare of hens kept in captivity [8,9], evidence shows that behaviors such as foraging, nest building, dustbathing and the search for elevated roosting areas are largely driven by internal factors, with the motivations to perform these behaviors best reduced via their performance [10]. Here we explore the burden of psychological pain associated with the deprivation of each of these behaviors.

Figure 6.1. Laying hens in a typical conventional cage used in commercial production. Illustration: Olga Kurkina.

Estimating Intensity

As discussed in Chapter 1 [5], the degree to which behavior and attention to other ongoing experiences are disrupted can be used as a yardstick to infer the intensity of painful sensations. As adaptive warning messages of actual or potential danger, the degree of unpleasantness associated with a negative affective state should also be related to the extent that failure to attend to it compromises the biological fitness of the individual, either as perceived or hardwired by natural selection [11]. In other words, the intensity of psychological pain should signify the corresponding degree of importance of the stimulus [12]. For instance, the rarer a critical resource (or the higher the cost to obtain it), the greater should be the unpleasantness associated with its lack or loss (e.g., the psychological distress associated with the severance of the maternal bond should be higher for mammals with heavy parental investment, as in such cases a stronger emotional bond to the offspring is favored by natural selection). Likewise, the more imminent a threat (e.g., the closer a predator), the greater should be the intensity of fear.

The classification used for the attribution of intensity to psychological harms is therefore the same as that used for physical harms (see Box 1.2 in Chapter 1 [5]). For example, psychological pain at the first (annoying) level should include experiences representing nuisances and discomfort that can be ignored most of the time, whereas at the second level (hurtful), the effect of drugs (in this case, drugs such as anxiolytics and antidepressants) on the alleviation of symptoms should be more easily observable. Psychological pain at the disabling level involves a higher degree of fear, anxiety, helplessness or despair that should prevent individuals from enjoying virtually any positive experience. The excruciating level represents the point of psychological breakdown, where self-harm and other maladaptive outbursts may be observed.

One rule of thumb to help estimate the intensity of a specific harm is to establish whether the affected individual trades the sensation produced by the harm for another with a better known intensity. For example, the pain associated with exposure to extreme cold can reach a disabling level (avoidance of extreme cold is critical for survival). Pain at the disabling level could be then attributed to the hunger sensation felt by individuals willing to endure that level of cold exposure to obtain food. The classifications of the intensity of the unpleasantness associated with negative affective states of psychological nature can be assisted by similar heuristics. For example, laying hens sustain multiple fractures that accumulate over the laying cycle, with clear evidence of chronic pain long after a fracture has healed (Chapter 3 [13]). Because behaviors such as jumps and short flights have been shown to be painful for fractured hens, the performance of these behaviors by fractured individuals would indicate their willingness to pay a cost (in terms of physical pain) for access to these behavioral opportunities (though there are subtleties that must be considered, such as the possible analgesic effect of engaging in distractions; Box 9.4 in Chapter 9 [14]). This approach makes minimal external assumptions about the importance of behavioral opportunities to animals, as it is based on the observation of their motivations and preferences.

A related experimental paradigm is widely employed in animal research to quantify the motivational value of different activities and resources: the measurement of demand elasticity as derived from behavioral demand functions. Demand elasticity is the degree to which the desire or motivation for something changes as its cost changes, and can be quantified by measuring the slope of the function relating the quantity of a resource (amount, time with access) to the cost an individual is prepared to pay for it (effort). Slopes that are shallower (less steep than 1.0) are described as inelastic (individuals are willing to pay high costs to access the resource), whereas those steeper than 1.0 are elastic (increasing costs decreases motivation to obtain the resource). Here, the demand for food is often used as a yardstick for assessing the value of other resources or to rank the slopes to produce a hierarchy of behavioral priorities [15]), thus helping establish how much animals value different behaviors and the degree to which their deprivation may affect them negatively.

Estimating Duration

Understanding the burden of psychological pain endured by laying hens housed in different contexts requires examining not only the intensity, but also the duration of the effects of thwarted behavioral needs; namely, understanding whether the frustration associated with the inability to perform behaviors like forage and move around is felt continuously, or if it is restricted temporally (e.g,. coinciding with the circadian rhythms that dictate the motivation to perform certain behaviors).

On one hand, it is well known that the intensity of some unpleasant feelings representing unmet physiological needs, such as hunger or thirst, increases over time; unmet psychosocial needs may behave similarly. For example, in dogs, the magnitude of the distress caused by social isolation and boredom is time-dependent, increasing over time [16]. In laying hens, there is evidence to suggest that this is also the case for some behaviors. Hens deprived of dustbathing material, or of the possibility to sham dustbathe [17], increase their dustbathing behavior when dustbathing is made possible again; some studies suggest that the longer the duration of the deprivation, the higher the amount of subsequent bathing behavior [18,19]. A similar 'rebound' pattern has been observed in hens deprived of the possibility to perform comfort behaviors, such as wing stretching and wing flapping, suggesting an increase in the motivational tendency to perform these behaviors with increased time of deprivation [20].

In other cases, behavioral deprivation is unlikely to exert its effect on the animal with a constant intensity over time, but rather to have an effect in bouts [21]. For example, it is possible that, while eating, a hen kept in a furnished cage does not find movement restriction aversive at that particular time, not until she is disrupted by a cage companion, or wants to move about, explore the environment or stretch her wings. In such cases, it makes more sense to think in terms of chronic intermittent stress [21], namely a succession of repeated stressors. Similarly, while the motivation to search for and build a nest has been shown to be extremely strong in the hours preceding oviposition [22], there is little evidence to suggest that it would be present after an egg has been laid.

It is also important to realize that the time spent in pain may not necessarily match the duration of the stressors immediately causing it. Chronic pain, whether physical or psychological, is predominantly maladaptive (not benefiting the survival or reproduction of the individual), and may be triggered by traumatic interventions. In the same way that debeaking can lead to the development of neuropathic pain, a chronic fearful state can also emerge from a trauma (e.g., transportation, handling, mutilation) at a sensitive moment during development.

In the next sections the potential negative effects emerging from the deprivation of highly motivated behaviors are discussed. Estimates of duration are provided in time endured in pain **per day** (daily time budget), with an assumed total of 280-420 days of experience (occurrences) in each housing environment (conventional cages, furnished cages and aviaries). The latter estimate assumes that flocks are depopulated at 60-80 weeks, and that the laying cycle starts at 20 weeks (the rearing phase is not considered). For each of the housing situations, we also explore the proportion of individuals in the flock affected by behavioral deprivation. The sensitivity of the final estimates to these assumptions can be tested at the interactive online tool available at http://www.pain-track.org/hens.

Deprivation of specific behavioral needs

Nest building

> "*The worst torture to which a battery hen is exposed is the inability to retire somewhere for the laying act. For the person who knows something about animals it is truly heart-rending to watch how a chicken tries again and again to crawl beneath her fellow-cage mates to search there in vain for cover*"
>
> Konrad Lorenz [23]

Nests are a highly valued resource for laying hens [10,24–28]. Driven by hormonal changes preceding ovulation, hens perform a sequence of behavioral patterns prior to oviposition that start with the search for a suitable nest site - a quiet and secluded location where they can safely incubate their clutch (Figure 6.2). Once a nest is selected, a sitting phase commences, in which the hen adopts a sitting posture, occasionally interspersed with short bouts of nest building activities, such as scratching the floor, rotating the body and collecting litter [29].

Figure 6.2. Laying hen incubating her clutch. Illustration: Olga Kurkina.

These behavioral patterns are under endocrine control, and can start as early as 100 min before oviposition [30], with most birds sitting for 25–40 min before oviposition. During this time, hens exhibit increased activity, restlessness and vocalizations [10,31,32]. Among hens kept in conventional cages, where they are denied a nest, these changes include unsettled behaviors that are reported to be symptomatic of frustration, such as stereotyped pacing, increased aggression and displacement preening [25,31]. For example, one study compared the pacing behavior of hens in cages with a nest box, without a nest box and with access to the nest box blocked when hens were 28, 32 and 36 weeks of age [33]. Hens with access to the nest box spent significantly less time pacing in the hour before egg laying compared with hens that had no experience of a nest box or those that had their nest box blocked, with no change in behavior as birds aged, suggesting that they did not adapt to the lack of a nest.

The high priority that hens attribute to finding and using a suitable nest has been demonstrated in a variety of other ways. Not only do most hens lay eggs in a nest box when provided with one [34,35], but considerable aggression has also been shown to occur outside nest boxes, indicating a high level of competition for the boxes [27]. In general, hens will pay high costs for access to a nest site, including walking past dominant or unfamiliar birds in narrow corridors, squeezing through narrow gaps and pushing through weighted doors [10,24,25]. As the sitting phase approaches, the costs paid to access a nest become increasingly higher [26]. In one experiment [28], the authors demonstrated that the motivation to access a nest increases as the oviposition draws near. In this experiment, ISA Brown hens were trained to push a door sealed with an electromagnet. Birds tested 20 min prior to oviposition had a work rate that was approximately three times higher than that of birdstested 40 min or more prior to oviposition, and significantly higher than the rate to access food after 4 hours of food deprivation. In earlier experiments, the price paid (in terms of work) to reach a suitable nest during the pre-laying phase was as high as that employed to obtain food after being food deprived for 28 hours [22].

These patterns show that the inability to find a nest is highly disruptive for the hen, with all attention drawn to this seeking drive in the moments prior to oviposition. At this time, hens are willing to pay high prices for access to a nest site. Likewise, the ability to conduct other routine activities is greatly impaired as sitting time approaches. Such a degree of disruption, along with the evolutionary value of finding suitable nests, indicates that the lack of a suitable site most likely leads to disabling feelings of frustration (see Box 1.2, Chapter 1 [5], for definitions of intensity), in line with the suggestion that the absence of a nest box is one the most severe sources of frustration for laying hens [24]. In Pain-Track 6.1, we assume that the intensity of frustration during the first half of the search phase is likely of a hurtful nature, increasing to a disabling nature during the second half (hence the 50% probability assigned to each category). The disabling nature of the pain is more prevalent during the sitting phase, as oviposition approaches, subsiding during oviposition. Here, we conservatively assume that the frustration associated with the lack of a nest is linked to the timing of nesting motivation prior to and during oviposition, but that it is completely absent after an egg has been laid.

Pain-Track 6.1. Hypothesis for the intensity and duration of the psychological pain endured per day as a result of nest building deprivation in laying hens. The estimated prevalence of birds affected by the deprivation in conventional cages, furnished cages and aviaries are respectively: 100%, 2-23% and 2-8% (see text for justification). The definition of each intensity category is provided in Box 1.2 (Chapter 1 [5]).

	i. Nest search	ii. Pre-oviposition sitting	iii. Oviposition and pos-oviposition sitting
Excruciating			
Disabling	**50%**	**80%**	**50%**
Hurtful	**50%**	**20%**	**50%**
Annoying			
	30-60 min	25-45 min	5-15 min

Pain-Track 6.1 applies to all hens deprived of a nest box. In the case of layers kept in furnished cages and cage-free aviaries, where a commercial nest box is provided, a fraction of individuals may still undergo the distress that lack of a suitable nesting site triggers. In these systems, a proportion of birds may lay their eggs on the floor (often referred to as 'floor layers'), even though a nest box is provided. We suggest that rather than indicating that these hens do not value the presence of a nest, floor laying reflects either exclusion of the nest box due to competition [27] (e.g., there may not be enough nest boxes, particularly if they all want to use them at the same time), and/or the outcome of some birds not finding the provided commercial nest boxes suitable. The latter possibility is suggested by the observation that floor layers are willing to pay high costs to keep looking for possible nests, even in the presence of a commercial box. In an experiment where hens had to squeeze through a narrow gap to move to an exploration pen where a nest box could be found, the birds continued to squeeze through the narrowest gaps to explore further, even when a commercial nest was made available [36]. Indeed, although most hens will accept different types of nests, they also show marked preferences for some types. In general, highly preferred sites are secluded and dark, containing some form of manipulable material [37]. Based on the average proportion of eggs laid outside the nest in each housing system, we assume that 2-8% of hens experience the frustration of not finding a suitable nest in cage-free aviaries [38–40] and 2-23% in furnished cages [41,42].

Perching and Roosting in Elevated Areas

Under natural conditions, free-living chickens seek elevated structures such as tree branches to perch on during the day [43] and to roost on at night [44] as a strategy against ground predators [45]. Elevated perches also allow hens to monitor the environment and escape from other hens. The motivation to seek elevation is particularly strong at night [44], and remains strong in commercial layer breeds. Hens of all currently studied breeds will perch at night and, to a lesser extent, during daytime [46]. The anti-predator function of accessing an elevated structure is supported by the finding that perch usage rates are higher in small groups, where individuals must be more vigilant [47].

The provision of perches in commercial systems can affect hen welfare in multiple ways beyond fulfilling laying hens' potential behavioral need to perch. On one hand, the use of perches have been shown to improve bone and muscle strength, reduce injurious pecking and reduce stocking density on the floor in aviary systems. On the other hand, perches may also lead to a higher incidence of keel bone fractures and vent pecking, depending on their structural properties, positioning and height [46]. In this chapter, our interest is only in the potential psychological pain caused by the deprivation of proper perching opportunities. The effects of fractures, injuries and other welfare harms are explored in other chapters.

As with other behavioral deprivations, the extent to which laying hens 'miss' absent perches they have never experienced remains to be determined. However, research suggests that roosting at night on an elevated perch remains a behavioral priority for modern layers, even in commercial indoor housing systems where they are protected from predation [35,46]. Several lines of evidence support this view. First, the overwhelming majority of hens use perches at night [48] when adequate perch space is available, which suggests that this is a highly motivated behavior. If roosting is prevented, hens show signs of unrest [49]. Further evidence comes from studies showing that hens will work to obtain access to a perch at night. Olsson and Keeling [50] increased the force required for a hen to open a weighted push-door to gain access to a perch from 25% of an individual's capacity to 100%, the maximum possible for that individual hen (calculated based on the force required to obtain food after 24 hours of food deprivation). The median resistance laying hens were willing to push was 75% of their maximum capacity, compared with 0% for a sham perch. These results indicate that laying hens assign a high priority to perching, a finding compatible with the expectation that behaviors favoring protection from predators (such as the preference to roost in elevated sites) should be under strong selection. Accordingly, there is also some evidence that laying hens provisioned with continuous access to elevated structures in free-range farms are less fearful than those deprived of aerial perches [51], likely as a result of the perceived reduction of the threat of predation (or aggression) when an animal is able to monitor their environment from a safe spot.

In light of these findings, we tentatively assume that the inability to use a perch to roost at night is distressing for hens kept in conventional cages. Specifically, we assume that the distress endured should be particularly intense during the time when the hen would be looking for a perch to roost; in general, about 1 h before sunset (or before lights are turned off), hens start perching for the night, a process that commonly takes about 30 to 60 minutes [44]. Given the high levels of motivation to access a perch, similar to those required to access food after 24 hours of deprivation, and the highly adaptive value of protection from predation, the intensity of the pain at this particular time is more likely of a hurtful nature. Since preference for perching is substantially stronger during the night, with a much less frequent use of perches during the day [44], we conservatively assume that the absence of perches during the day is not a major driver of psychological pain in commercial layers. Such an assumption also avoids inflating the psychological pain endured by caged hens during active hours, at a time when we assume they might also endure the pain associated with deprivation of other behavioral needs, such as foraging and nesting.

Unfortunately, data on the behavior (or other welfare indicators) of caged layers during the night compared to that of hens who have the opportunity to spend the night on aerial perches is scant. Whether the sleep and vigilance patterns of caged hens are in any way altered when they are forced to sleep in locations potentially perceived as less safe remains to be determined. Blokhuis [44] observed that standing during the evening occurred more frequently in alarming situations, though it is not clear whether such a disruption is also present under typical commercial conditions or detrimental to hen welfare. Chickens have an additional phase of sleep that humans lack: unihemispheric slow-wave sleep, during which one half of the brain rests while the other half is awake;this is why chickens can sleep with one eye open and the other eye closed, an adaptation that allows them to be concurrently alert for predators and asleep [52]. For example, on average, hens spend 75% of dark hours in a sleeping posture, 7% active and 18% motionless with at least one eye open [53], during which they may exhibit fully developed large amplitude slow waves characteristic of electrophysiological sleep. Although unihemispheric sleep would mean a reduction of the time spent sleeping compared to bihemispheric sleep, the behavior and health of birds does not seem to be impaired by a reduction of sleep [52]. Therefore, we conservatively assume that a lack of elevated structures upon which to roost at night has a low likelihood of being mildly distressing to hens in conventional cages (Pain-Track 6.2), during 5-15% of sleeping time. Further research is needed to address the possibilities raised here.

Pain-Track 6.2. Hypothesis for the intensity and duration of the psychological pain endured per day by commercial layers as a result of deprivation of an elevated site for roosting in conventional cages. The estimated prevalence of birds affected is 100% (see Box 1.2, Chapter 1 [5], for the definition of each intensity category).

	i. Search for safe, elevated site	ii. Dark hours
Excruciating		
Disabling		
Hurtful	80%	5%
Annoying	20%	10%
	30-60 min	6-8 hours

In furnished cages, perches are usually placed in two or more parallel rows along the length of the cage, but the cage height limits the perch height, and normal hen movement within the cage may disrupt perching [46]. In these cages, about 80–95% of hens have been shown to roost on perches at night [10], even though evidence suggests that hens strongly favor more elevated areas relative to those perceived to be close to the ground [46]. Therefore, we assume that the inability to use an elevated location to roost during the evening is still distressing for hens kept in furnished cages, although the presence of a perch partially satisfies this need. The likelihood of experiencing higher levels of distress is therefore reduced in Pain-Track 6.3.

Pain-Track 6.3. Hypothesis for the intensity and duration of the psychological pain endured per day by commercial layers as a result of deprivation of an elevated site for roosting in furnished cages. The estimated prevalence of birds affected is 100%. The definition of intensity categories is provided in Box 1.2 (Chapter 1 [5]).

	i. Search for safe, elevated site	ii. Dark hours
Excruciating		
Disabling		
Hurtful	50%	
Annoying	50%	15%
	30-60 min	6-8 hours

In cage-free aviaries (Pain-Track 6.4), perch use is high, especially at night, when up to 90% of hens are observed roosting on the top-level perch [27,54]. In line with this observation, we assume that 75-95% of hens in aviaries are able to find a suitable elevated perch to roost at night. The remaining birds (5-25%) are assumed to use lower level perches. In such cases, the level of distress endured is conservatively assumed to be similar to that experienced by perching hens in furnished cages.

Pain-Track 6.4. Hypothesis for the intensity and duration of the psychological pain endured per day by commercial layers as a result of deprivation of an elevated site for roosting in cage-free aviaries. The estimated prevalence of birds affected ranges between 5 to 25% (see text for justification).

	i. Search for safe, elevated site	ii. Dark hours
Excruciating		
Disabling		
Hurtful	50%	
Annoying	50%	15%
	30-60 min	6-8 hours

Foraging and Exploration

To insure against the risk of starvation associated with the unpredictable availability of food, animals must constantly gather information about the presence, nature and location of alternative food sources. Foraging, namely the exploration of the environment for the (immediate or future) acquisition of food, is one of the most important elements in the behavioral repertoire of animals, particularly omnivorous species such as chickens. According to the "information hypothesis", investing energy to gather information about potential food sources pays off by increasing the likelihood of survival when individuals are faced with fluctuations in food availability [55]. In this section we explore the possible psychological pain emerging from the deprivation of foraging and of the possibility of exploring the environment; since they are so closely intertwined, we explore the degree of frustration emerging from the deprivation of these behaviors jointly.

In a study that observed the behavior of junglefowl in captive conditions [56], the proportion of the day spent foraging for food dominated the birds' time budgets, with 60% of active time spent pecking the ground (even though the birds were regularly fed) and 34% spent scratching the ground. Among commercial layers in cage-free settings, a large proportion of the active hours remain dedicated to foraging (from 25% to around 40% of daytime [10,57]) and moving about, even though the environment is much more predictable, and selection for high productivity diverts a larger share of energy to the metabolic processes associated with egg production.

Accordingly, there is evidence to suggest that modern layer breeds still exhibit a high demand for a litter substrate, consistently working to access it [10,58–60], even in the presence of freely available food (a phenomenon known as 'contrafreeloading') and in the absence of food rewards within the substrate. It is sometimes difficult to distinguish whether hens are motivated to gain access to litter in order to forage or dustbathe (they can perform both behaviors in some litter substrates such as sand) [61]; however, experiments aimed at establishing the cost layers were prepared to pay for the opportunity to access foraging substrates revealed that demand for substrates such as straw, which is not used for dustbathing, remains very high and inelastic [58] and similar to that observed for food. In these experiments, the imposition of a cost did not alter the amount of time spent pecking and scratching, suggesting that foraging remains a behavioral need in modern commercial layer strains.

Observations that hens continue to forage and move around after sustaining multiple fractures known to be painful (Chapter 3 [13]) indicate their willingness to pay a cost, in terms of physical pain, to be able to perform these behaviors (though it is possible that, while distracted, the salience of physical pain is reduced; Chapter 9 [14]). In video observations of laying hens kept in aviaries, the mean number of hens in the open litter area was similar at the end of lay (when the pain associated with fractures was likely worse) compared to mid or peak lay [62]. In another study, up to 85% of hens were simultaneously observed on the litter shavings at 68 weeks of age, a higher percentage than that observed at 28 weeks of age [63].

Other lines of evidence also suggest that foraging deprivation is detrimental to the well-being of layers. For example, aggressive pecking has been shown to be more frequent if litter space is insufficient [27]. Lower baseline levels of corticosterone (a marker of physiological arousal) are also observed in layers who choose areas with additional foraging opportunities, compared to those who choose environments devoid of such opportunities [64,65]. Additionally, differences in the degree of exploratory behavior of layers deprived of litter for six weeks have been also taken to indicate that they ‘miss’ access to foraging opportunities during the deprivation period, as opposed to simply working to obtain access to these opportunities when they are made available [10,66]. Finally, when given a choice between ‘the lesser of two evils’, hen’s preferences suggest that being deprived of the opportunity to forage and explore the environment is the worst of them. For example, even though caged hens prefer larger cages [67]), they have been shown to prefer a tiny cage (0.38 x 0.215 m), which prevented even the most basic movements such as turning around, with litter on the floor more than a much larger cage (0.76 x 0.86 m) with a wire floor [68]. These observations were consistent even after hens had experience with the cramped conditions of the tiny cage (ruling out the possibility that they were simply attracted to the litter when they first saw it), indicating that hens assign a higher priority to litter than to space, at least over the short term.

By contrasting these observations with our definitions of pain intensity (Box 1.2, Chapter 1 [5]), some inferences are possible. First, because hens deprived of the ability to forage do not refrain from activities such as eating, drinking, and reacting to external stimuli, the degree to which foraging deprivation affects them is unlikely to be of a disabling (or more intense) nature. Conversely, the observation that hens are motivated to pay a high energetic cost for access to foraging opportunities (similar to that paid for access to food) suggests that this deprivation is more than an annoyance to the animal, at least one that cannot be ignored most of the time. The fact that fractured hens keep visiting the litter area lends support to this possibility. Indeed, of the core instincts identified in mammalian brains (anger, fear, panic-grief, maternal care, pleasure/lust, play, and seeking), seeking is the most important [12], and there is no evolutionary reason to expect that this selective pressure would not be as strong in birds [69]. However, given the lack of a preference for foraging substrates over wire under some circumstances [70], we conservatively assume that the inability to forage and explore the environment is only associated with discomfort in some cases (hence the probability assigned to the annoying category in the pain-tracks). These probabilities can also be interpreted as representing fluctuations in the level of distress over a day and over the laying cycle.

In terms of duration, the extent to which this frustration is sustained over an entire day and laying cycle is not clear. On one hand, it is possible that the drive to forage and gather information is only interrupted when other, more immediate, needs are present [61]; the fact that wild junglefowl dedicate about 90% of their time to foraging and exploration supports this expectation. Unlike other behaviors clearly associated with a circadian rhythm (e.g., nesting), the frustration emerging from the thwarted motivation to forage and explore may thus be present at all times, with the exception of the time the hen's attention is focused on activities such as feeding, drinking, nesting and dustbathing; other than when engaged in these behaviors, hens offered the opportunity to forage will majoritarily take it. Therefore, it would not be unreasonable to assume that hens would be frustrated for approximately 60-80% of active hours, or for approximately 8 to 12 hours given a typical light:dark regime of 16:8 hours. On the other hand, we cannot rule out the possibility that psychological pain is restricted to the time periods when a commercially-raised hen would be foraging given the availability of proper resources (i.e., space, light, litter), with it being completely absent at all other times. Assuming that 25-40% of the active daytime is dedicated to foraging and exploration, this translates into approximately 4 to 6.4 hours per day. Uncertainty regarding the more appropriate hypothesis is represented by the wide uncertainty intervals in pain-tracks 6.5 to 6.7, ranging from 4 to 12 hours.

Foraging and exploration are not possible in conventional cages (Pain-Track 6.5). In furnished cages, these behaviors are only minimally accommodated. The scratch pad provided to encourage foraging quickly becomes depleted of any litter and contaminated with manure. Accordingly, the time budget allocated to behaviors such as ground pecking is significantly reduced compared to cage-free housing [71]. Additionally, the extent to which a lack of space, litter, positive feedback from foraging and novelty add to the frustration derived from the inability to perform typical foraging activities in furnished cages is currently unknown.

Pain-Track 6.5. Hypothesis for the intensity and duration of the psychological pain endured per day by commercial layers as a result of the deprivation to forage and explore the environment in conventional cages. The estimated prevalence of birds affected is 100%.

	i. Deprivation of foraging bouts
Excruciating	
Disabling	
Hurtful	**80%**
Annoying	**20%**
	4 - 12 hours

In Pain-Track 6.6 we assume that the distress associated with the thwarting of foraging behavior in furnished cages is only slightly improved relative to conventional cages, represented by the higher likelihood that the pain is only annoying.

Pain-Track 6.6. Hypothesis for the intensity and duration of the psychological pain endured per day by commercial layers as a result of the deprivation to forage and explore the environment in furnished cages. The estimated prevalence of birds affected is 100%

	i. Deprivation of foraging bouts
Excruciating	
Disabling	
Hurtful	**60%**
Annoying	**40%**
	4 - 12 hours

Foraging and exploration needs are better fulfilled in cage-free aviaries, provided that sufficient space, lighting levels and adequate litter materials are available. In these circumstances, no loss of welfare arising from the deprivation of foraging would be expected. However, some factors may impair the full expression of behavioral opportunities by all birds in cage-free aviaries. Even though the extent of potential deprivations is small relative to a caged environment (where virtually all behavioral needs are thwarted), they must still be considered. In addition to high stocking densities and the lack of proper litter material, movement may be restricted by the use of divisions between units or limited points of entry between the litter and the aviary tiers [62,72]. In addition, factors such as social facilitation and circadian rhythms may lead to the temporal aggregation of individuals in the litter area, which may then become insufficient to accommodate all hens [62,72]. Consequently, despite having more freedom of movement, some hens may not fully experience all the intended foraging benefits of cage-free environments. Additionally, foraging may be constrained if the health of the hen is somehow impaired. As discussed in previous chapters, layers are likely to experience painful reproductive disorders (Chapter 5 [73]) and bone fractures (Chapter 3 [13]) that may accumulate over the laying cycle, so that by the end of lay their motivation to forage and explore the environment may be reduced as a by-product of physical pain. In a longitudinal study that followed focal birds over an entire laying cycle, there were no significant differences in the average time allocated to foraging as the hens aged, though the maximum and minimum times spent foraging were longer in the beginning of the cycle [74]. Understanding the extent to which animals can fulfil their behavioral needs in cage-free facilities is critical, so further research is very much needed to enable an estimation of not only the proportion of animals within a flock that do not have access to important behavioral opportunities, but also the proportion of cage-free houses that provide hens with adequate resources for their fulfillment. Here, we estimate that 5 to 20% of hens in cage-free aviaries are incapable of fulfilling their foraging needs. This range encompasses both our uncertainty regarding this proportion, as well as the natural variability expected across genetic breeds [72] and management conditions. This estimate can also be interpreted as the average proportion of time that hens are unable to forage (when they were supposed to), despite a strong motivation to engage in this behavior. In such cases (Pain-Track 6.7), we assume that the distress associated with the thwarting of foraging behavior is slightly improved relative to furnished cages, given the higher freedom of movement in cage-free conditions.

Pain-Track 6.7. Hypothesis for the intensity and duration of the psychological pain endured per day by commercial layers as a result of the deprivation to forage and explore the environment in aviaries. The estimated prevalence of birds affected ranges between 5 and 20% (see text for justification).

	i. Deprivation of foraging bouts
Excruciating	
Disabling	
Hurtful	**40%**
Annoying	**60%**
	4 - 12 hours

Dustbathing

Dustbathing, which is common in poultry and other bird species, consists of a sequence of coordinated movements. Initially, the bird squats in the substrate and uses their wings to throw dust through their feathers, next rising and shaking off the dust. Other elements include raking of the substrate with the beak, ground scratching, wing shaking, and head and side rubbing [75]. From a functional perspective, dustbathing has been associated with the removal of surplus stale lipids [76] (improving the insulation capacity of the feathers), dandruff and possibly ectoparasites [77] from the feathers, although the latter function is still controversial [78].

In laying hens, dustbathing seems to be regulated by interactions between external factors (e.g., the sight of a suitable substrate [79], light, heat) and internal drivers, with a clear diurnal rhythm [80]. In general, dustbathing occurs more frequently in the middle of the day and afternoon in systems in which hens have continuous access to appropriate substrates. When given access to soil, sand or litter, hens will dustbathe about once every other day, often for 20 to 35 minutes at a time [80].

Laying hens kept in conventional cage systems are unable to dustbathe. Nonetheless, despite the lack of space and litter, caged hens will try to perform the behavioral elements of dustbathing, a process often referred to as ‘sham’ dustbathing. Sham dustbathing lacks positive feedback for the hen [81,82], involving an incomplete behavioral sequence that does not seem to reduce the hen’s motivation to dustbathe [83].

Hens have been shown to be more motivated to dustbathe after a period of deprivation; overall, the longer the deprivation, the higher the amount of dustbathing behavior that is performed when the opportunity arises [18,19,80]. Similarly, caged hens deprived of a litter area have been shown to be more motivated to dustbathe than aviary-housed hens: their latency to dustbathe is shorter, dust baths were longer and more numerous and more hens dust-bathed among caged hens than among aviary hens [84]. These findings suggest an increase in motivation after deprivation, hence the influence of internal factors [76], although the possibility that deprived hens are simply reacting to the novelty of the presence of the dustbathing material has also been considered [20]. Additionally, the removal of dustbathing materials has been shown to elicit gakel-calls [85] and increase exploratory behavior [66], suggesting that layers 'miss' this behavioral opportunity. This possibility is also supported by observations showing a higher number of stereotypic pecks performed by birds kept in wire cages compared to caged birds with access to sand [19]. Moreover, the low performance of dustbathing in red junglefowl has been associated with increased fear [86], although causality cannot be inferred from these findings. Notwithstanding, a number of studies has failed to produce evidence of a motivation to dustbathe [87–90]. This has led to suggestions that dustbathing could be of a lesser priority than other behaviors [35,91], with some authors proposing that rather than a need, dustbathing may be a pleasurable activity that layers engage in when given the opportunity [81].

In light of the conflicting evidence as to whether the deprivation of proper dustbathing opportunities has a negative impact on welfare, we tentatively assign a 50% probability that the welfare of hens is not negatively impacted in any way, and similarly a 50% probability that there is some frustration associated with deprivation, predominantly of an annoying nature (Pain-Track 6.8). In the latter case, observations that dustbathing increases with time since its last performance, with compensatory adjustments when the amount of previous dustbathing is reduced [18,19,80,92], indicate that the putative need to dustbathe during periods of deprivation may accumulate over time. Thus, we consider that there is some chance that the annoyance increases to a more intense level. These probabilities can also be interpreted as a temporal distribution of frustration levels in the intensity categories, with 35% of the time spent at the annoying level, 15% at the hurtful, and 50% of the interval with no distress. In terms of duration, we conservatively assume that frustration would be present during the hours typically associated with the performance of the behavior, namely in the middle of the day and afternoon (from approximately 11:00 to 16:00). In addition to this five-hour interval, we add the possibility of a 50% shorter or 50% longer interval (hence ranging from 2.5 to 7.5 hours), given the high level of uncertainty associated with these assumptions.

Pain-Track 6.8. Hypothesis for the intensity and duration of the psychological pain endured per day by commercial layers as a result of dustbathing deprivation in conventional cages. The estimated prevalence of birds affected is 100%.

	i. Deprivation of dustbathing
Excruciating	
Disabling	
Hurtful	**15%**
Annoying	**35%**
	2.5 - 7.5 hours

Dustbathing is not properly accommodated in furnished cages [93], where sham dustbathing is often observed [94]. In general, any material that is added to plastic turf mats (Astroturf) placed in the cage to allow for foraging and dustbathing is rapidly depleted; this is particularly the case given the frequent use of food particles as litter substrate [95]. Additionally, only one hen can dustbathe on these mats at a time, which may cause frustration in the other hens given the synchronous nature of this behavior [96]. Also, substrates with fine particles such as peat and sand are preferred over Astroturf, and behavior on less preferred substrates is generally more unsettled, with shorter bouts that may indicate some degree of frustration [97]. Pain-Track 6.9 thus considers the possibility that dustbathing in furnished cages may also be frustrating for most laying hens, although less so (hence a lower pain intensity) than in conventional cages .

Pain-Track 6.9. Hypothesis for the intensity and duration of the psychological pain endured per day by commercial layers as a result of dustbathing deprivation in furnished cages. The estimated prevalence of birds affected is 100%

	i. Deprivation of dustbathing
Excruciating	
Disabling	
Hurtful	5%
Annoying	45%
	2.5 - 7.5 hours

In cage-free aviaries, opportunities for dustbathing are not constrained as long as the appropriate substrates are provided and the space available is sufficient for the hens to all simultaneously dustbathe, considering the synchronicity of the behavior. Naturally, in aviaries with only plastic or wire floors, or where litter is non-friable, shallow or becomes too wet, proper dustbathing is not possible [63]. Given the high level of heterogeneity in management practices, and uncertainty surrounding this parameter, we assume that anywhere from 10 to 50% of hens will be unable to properly dustbathe, when motivated to do so, in indoor aviary systems. Because of their greater freedom of movement, the intensity distribution in such cases is slightly milder than in furnished cages (Pain-Track 6.10).

Pain-Track 6.10. Hypothesis for the intensity and duration of the psychological pain endured per day by commercial layers as a result of dustbathing deprivation in conventional cages. The estimated prevalence of birds affected ranges between 10 and 50% (see text for justification)

	i. Deprivation of dustbathing
Excruciating	
Disabling	
Hurtful	
Annoying	50%
	2.5 - 7.5 hours

Movement restriction

Hens need space to stretch and flap their wings, preen their feathers, shake their bodies and move around. In conventional cage systems, and to a greater extent in furnished cages, laying hens do not have enough space to engage in these behaviors, or enough space to even adopt postures (e.g., standing posture with the head erect) that are typical of their behavioral repertoire [98]. Space and movement restriction is so severe that individuals spend most of their time spatially restrained by direct physical contact with other birds (Figure 6.3).

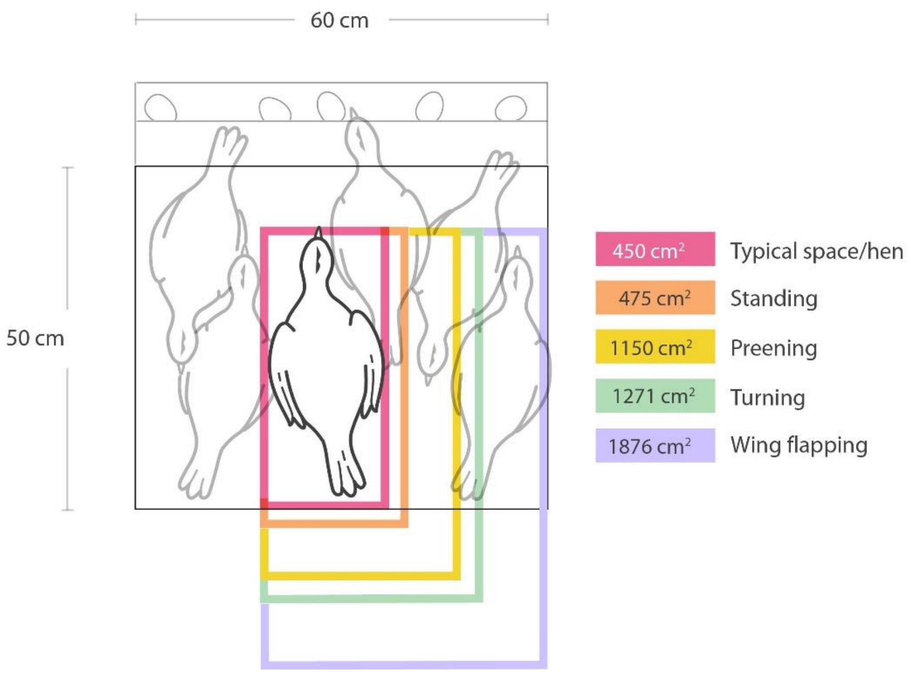

Figure 6.3. Area required to perform several comfort behaviors relative to the space typically available in conventional cage systems. Adapted from Dawkins and Hardie [99]. Illustration: Alberto Gareppe.

The inability to move around, perform comforting behaviors and adopt typical postures most likely leads to a permanent feeling of discomfort. It has indeed been suggested that such constraints lead to frustration [20,100]. Here, we assume that hens in conventional cages are likely to endure a discomforting feeling (only of an annoying nature) during the 14-16 hours of light typically employed in commercial systems (Pain-Track 6.11). However, we subtract the time spent feeding (2-4 hours) from this interval, to a total of 10-14 hours endured with discomfort due to crowding and the severe restriction of movement. In furnished cages, this discomfort is somewhat reduced given the greater availability of space (600 square centimeters per hen), though this space is clearly still not sufficient for the performance of many behavioral patterns (Figure 6.3); we tentatively assume a 40% reduction in discomfort in hens kept in this system (Pain-Track 6.12). In aviaries, this sustained state of movement restriction is not present, hence no pain-track was developed for this housing type.

Pain-Track 6.11. Hypothesis for the intensity and duration of the psychological pain endured per day by commercial layers as a result of discomfort due to movement restriction in conventional cages. The estimated prevalence of birds affected is 100%.

	i. Discomfort due to movement restriction
Excruciating	
Disabling	
Hurtful	
Annoying	**100%**
	12 - 14 Hours

Pain-Track 6.12. Hypothesis for the intensity and duration of the psychological pain endured per day by commercial layers as a result of discomfort due to movement restriction in furnished cages. The estimated prevalence of birds affected is 100%.

	i. Discomfort due to movement restriction
Excruciating	
Disabling	
Hurtful	
Annoying	**60%**
	12 - 14 hours

Burden of Pain

Here we quantify the time spent in psychological pain (in hours) as a result of the deprivation of various motivated behaviors as experienced by laying hens raised commercially. Figure 6.4 shows the expected time endured at each level of pain intensity (hours), per bird affected, as a result of the deprivation of each behavior in each housing system. In the case of cage-free aviaries, the chart depicts the expected time in pain for an individual that is prevented from expressing the behavior, either due to a lack of conditions or poor health. In all cases, estimated times are based on the pain-tracks presented and justified previously. Only hours spent awake (16 hours/day) are considered. The results presented in Figure 6.4 consider the uncertainty regarding the intensity of pain and duration of each time segment in the pain-tracks, propagated through the model. However, if the reader wants to change any of the assumptions regarding the durations and pain intensities, an interactive web application can be used for this purpose, available at https://pain-track.org/hens.

Figure 6.4. Time endured at each level of pain intensity (hours), per bird affected, as a result of the deprivation of behaviors that laying hens are highly motivated to perform. Note the different time scales for different intensities of pain. In the case of cage-free aviaries, the chart depicts the expected time in pain for an individual that is prevented from expressing the behavior, either due to a lack of conditions or poor health. Error bars represent 95% confidence intervals. An interactive comparison with other harms is available at https://www.hen-welfare.org/charts.

Although the frustration emerging from the inability to find a nest in which to lay an egg is shorter than the inability to perform other motivated behaviors (given the circadian nature of the behavior), it leads to an average of 324 hours of disabling pain over one laying cycle (or nearly one hour per day), more than any other source of disabling pain analysed in this book. In turn, the inability to forage and seek resources/novelty underlies the longest time in frustration at the Hurtful level, a finding naturally emerging from the observation that most of the daily time budget of hens is dedicated to this activity. The inability to dustbathe, roost in elevated areas and, especially, to move and perform a series of comfort behaviors (Figure 6.3) add to this burden, leading to long hours of discomfort (pain at the annoying level) in conventional cages, and to a lower extent in furnished cages.

Figure 6.5. Beak-trimmed laying hen. Illustration: Olga Kurkina.

Research Gaps

- ❏ The extent to which behavioral opportunities are realized in cage-free systems must be better established in terms of (1) the conditions offered (e.g., surveys determining the proportion of facilities that offer hens proper litter material, space and lighting conditions) and (2) the extent to which the hens make use of them. To this end, the entire laying cycle must be considered (e.g., behavior may be constrained if the health of the hen is somehow impaired).
- ❏ Data on the behavior (and other welfare indicators) of layers during the night is scant. It remains unknown whether the sleep and vigilance patterns of caged hens are in any way altered when they are forced to sleep in locations potentially perceived as less safe .
- ❏ Currently, little is known about the evolution of feelings of frustration (associated with the thwarted motivation to forage and explore the environment) with time (e.g., whether it accumulates , or remains unchanged, over the laying cycle). Research on motivational preferences, coupled with ethological observations of caged hens over the laying cycle, controlling for differences in health, egg production and feed consumption, could shed light on this question.
- ❏ Evidence on the extent to which the deprivation of proper dustbathing opportunities has a negative impact on welfare is conflicting. Little is known about the duration and evolution of its adverse (psychological) effects over the day and the laying cycle.

References

1. West B, Zhou B-X. Did chickens go North? New evidence for domestication. J Archaeol Sci. 1988;15: 515–533.

2. Broom DM. A history of animal welfare science. Acta Biotheor. 2011;59: 121–137.

3. Nicol CJ. Feather Pecking in Laying Hens: Why They Do It, and Welfare Implications. Poultry Feathers and Skin: The Poultry Integument in Health and Welfare. 2019; 31.

4. IASP. IASP Announces Revised Definition of Pain. 16 Jul 2020 [cited 13 Aug 2020].

5. Alonso WJ, Schuck-Paim C. The Comparative Measurement of Welfare in Animals: Cumulative Time in Pain as a Universal Metric with Biological Meaning. In: Schuck-Paim C, Alonso WJ, editors. Quantifying Pain in Laying Hens. Independently published; 2021.

6. Schuck-Paim C, Alonso WJ. Welfare Implications of Injurious Pecking in Laying Hens. In: Schuck-Paim C, Alonso WJ, editors. Quantifying Pain in Laying Hens. Independently published; 2021.

7. Umar S, Munir MT, Ahsan U, Raza I, Chowdhury MR, Ahmed Z, et al. Immunosuppressive interactions of viral diseases in poultry. Worlds Poult Sci J. 2017;73: 121–135.

8. Dawkins MS. Behavioural deprivation: A central problem in animal welfare. Appl Anim Behav Sci. 1988;20: 209–225.

9. Mason G, Burn C. Behavioural Restriction. In: Appleby, editor. Animal Welfare. CAB International; 2011. p. 98.

10. Nicol C. The Behavioural Biology of Chickens. CAB International; 2015.

11. Fraser D, Duncan IJH. “Pleasures”,’Pains' and Animal Welfare: Toward a Natural History of Affect. Anim Welf. 1998;7: 383–396.

12. Panksepp J. Affective Neuroscience: The Foundations of Human and Animal Emotions (Series in Affective Science). 1 edition. Oxford University Press; 2004.

13. Schuck-Paim C, Alonso WJ. Quantifying the Pain due to Keel Bone Fractures in Laying Hens. In: Schuck-Paim C, Alonso WJ, editors. Quantifying Pain in Laying Hens. Independently published; 2021.

14. Schuck-Paim C, Alonso WJ. Impact of the Transition from Caged to Cage-free Housing on the Welfare of Laying Hens. In: Schuck-Paim C, Alonso WJ, editors. Quantifying Pain in Laying Hens. Independently published; 2021.

15. Weeks CA, Nicol CJ. Laywel: welfare implications of changes in production systems for laying hens. Deliverable 7.1 Overall strengths and weaknesses of each defined housing system for laying hens, and detailing the overall welfare impact of each housing system. 2007.

16. McMillan FD. Maximizing quality of life in ill animals. J Am Anim Hosp Assoc. 2003;39: 227–235.

17. van Liere DW, Wiepkema PR. Effects of long-term deprivation of sand on dustbathing behaviour in laying hens. Anim Behav. 1992;43: 549–558.

18. Vestergaard K. The Regulation of Dustbathing and Other Behaviour Patterns in the Laying Hen: A Lorenzian Approach. In: Moss R, editor. The Laying Hen and its Environment: A Seminar in the EEC Programme of Coordination of Research on Animal Welfare, organised by R Moss and V Fischbach, and held at Luxembourg, March 11–13, 1980. Dordrecht: Springer Netherlands; 1980. pp. 101–120.

19. Vestergaard KS, Skadhauge E, Lawson LG. The stress of not being able to perform dustbathing in laying hens. Physiol Behav. 1997;62: 413–419.

20. Nicol CJ. Behavioural responses of laying hens following a period of spatial restriction. Anim Behav. 1987;35: 1709–1719.

21. Ladewig J. Chronic intermittent stress: a model for the study of long-term stressors. In: Moberg GP, Mench JA, editors. The biology of animal stress: basic principles and implications for animal welfare. CAB International; 2000. p. 159.

22. Follensbee ME, Duncan IJH, Widowski TM. Quantifying nesting motivation of domestic hens. J Anim Sci. 1992;70.

23. Lorenz K. Animals are sentient beings: Konrad Lorenz on instinct and modern factory farming. Devendran S, trans Der Spiegel. 1980;34: 251–264.

24. Duncan IJH. The pros and cons of cages. Worlds Poult Sci J. 2001;57: 381–390.

25. Duncan IJH. The Philosophical and Biological Evolution of Feelings in Well-being. In: McMillan FD, editor. Mental Health and Well-being in Animals. CAB International; 2020. pp. 21–32.

26. Freire R, Appleby MC, Hughes BO. Effects of nest quality and other cues for exploration on pre-laying behaviour. Appl Anim Behav Sci. 1996;48: 37–46.

27. Odén K, Keeling LJ, Algers B. Behaviour of laying hens in two types of aviary systems on 25 commercial farms in Sweden. Br Poult Sci. 2002;43: 169–181.

28. Cooper JJ, Albentosa MJ. Behavioural Priorities of Laying Hens. Avian and Poultry Biology Reviews. 2003;14: 127–149.

29. Cronin GM, the late John L. Barnett, Hemsworth PH. The importance of pre-laying behaviour and nest boxes for laying hen welfare: a review. Anim Produc Sci. 2012;52: 398–405.

30. Sherwin CM, Nicol CJ. A descriptive account of the pre-laying behaviour of hens housed individually in modified cages with nests. Appl Anim Behav Sci. 1993;38: 49–60.

31. Meijsser FM, Hughes BO. Comparative analysis of pre-laying behaviour in battery cages and in three alternative systems. null. 1989;30: 747–760.

32. Wood-Gush DG, Gilbert AB. Oestrogen and the pre-laying behaviour of the domestic hen. Anim Behav. 1969;17: 586–589.

33. Yue S, Duncan IJH. Frustrated nesting behaviour: relation to extra-cuticular shell calcium and bone strength in White Leghorn hens. Br Poult Sci. 2003;44: 175–181.

34. Appleby MC, Walker AW, Nicol CJ, Lindberg AC, Freire R, Hughes BO, et al. Development of furnished cages for laying hens. Br Poult Sci. 2002;43: 489–500.

35. Weeks CA, Nicol CJ. Behavioural needs, priorities and preferences of laying hens. Worlds Poult Sci J. 2006;62.

36. Cooper JJ, Appleby MC. Motivational aspects of individual variation in response to nestboxes by laying hens. Anim Behav. 1997;54: 1245–1253.

37. Clausen T, Riber AB. Effect of heterogeneity of nest boxes on occurrence of gregarious nesting in laying hens. Appl Anim Behav Sci. 2012;142: 168–175.

38. Icken W, Thurner S, Heinrich A, Kaiser A, Cavero D, Wendl G, et al. Higher precision level at individual laying performance tests in noncage housing systems. Poult Sci. 2013;92: 2276–2282.

39. Oliveira JL, Xin H, Chai L, Millman ST. Effects of litter floor access and inclusion of experienced hens in aviary housing on floor eggs, litter condition, air quality, and hen welfare. Poult Sci. 2019;98: 1664–1677.

40. Sørensen P, Okeno T, Buitenhuis AJ. The motivation of hens to lay eggs on the floor in non-cage systems has a heritable background. EuropPoultSci. 2017;81.

41. Hunniford ME, Torrey S, Bédécarrats G, Duncan IJH, Widowski TM. Evidence of competition for nest sites by laying hens in large furnished cages. Appl Anim Behav Sci. 2014;161: 95–104.

42. Cepero R, Hernandiz A. Effects of housing systems for laying hens on egg quality and safety. XVI European Symposium on the Quality of Eggs and Egg Products. 2015.

43. Duncan IJH, Savory CJ, Wood-Gush DGM. Observations on the reproductive behaviour of domestic fowl in the wild. Applied Animal Ethology. 1978;4: 29–42.

44. Blokhuis HJ. Rest in poultry. Appl Anim Behav Sci. 1984;12: 289–303.

45. Jordania J. Why Do People Sing? Music in Human Evolution. First edition. Tbilisi: Logos; 2011.

46. O'Connor A, Dzikamunhenga RS, Totton S, Wolfe D, Sargeant J, Glanville J, et al. Systematic review of the effect of perch height on keel bone fractures, deformation and injuries, bone strength, foot lesions and perching behavior. EFSA Supporting Publications. 2015;12: 841E.

47. Newberry RC, Estevez I, Keeling LJ. Group size and perching behaviour in young domestic fowl. Appl Anim Behav Sci. 2001;73: 117–129.

48. Braastad BO. Effects on behaviour and plumage of a key-stimuli floor and a perch in triple cages for laying hens. Appl Anim Behav Sci. 1990;27: 127–139.

49. Olsson IA, Keeling LJ. Night-time roosting in laying hens and the effect of thwarting access to perches. Appl Anim Behav Sci. 2000;68: 243–256.

50. Olsson IAS, Keeling LJ. The Push-Door for Measuring Motivation in Hens: Laying Hens are Motivated to Perch at Night. Anim Welf. 2002;11: 11–19.

51. Donaldson CJ, O'Connell NE. The influence of access to aerial perches on fearfulness, social behaviour and production parameters in free-range laying hens. Appl Anim Behav Sci. 2012;142: 51–60.

52. Mascetti GG. Unihemispheric sleep and asymmetrical sleep: behavioral, neurophysiological, and functional perspectives. Nat Sci Sleep. 2016;8: 221–238.

53. van Luijtelaar EL, van der Grinten CP, Blokhuis HJ, Coenen AM. Sleep in the domestic hen (Gallus domesticus). Physiol Behav. 1987;41: 409–414.

54. Abrahamsson P, Tauson R. Aviary Systems and Conventional Cages for Laying Hens: Effects on Production, Egg Quality, Health and Bird Location in Three Hybrids. Acta Agric Scand A Anim Sci. 1995;45: 191–203.

55. Inglis IR, Ferguson NJK. Starlings search for food rather than eat freely-available, identical food. Anim Behav. 1986;34: 614–617.

56. Dawkins MS. Time budgets in Red Junglefowl as a baseline for the assessment of welfare in domestic fowl. Appl Anim Behav Sci. 1989;24: 77–80.

57. Schutz KE, Jensen P. Effects of Resource Allocation on Behavioural Strategies: A Comparison of Red Junglefowl (Gallus gallus) and Two Domesticated Breeds of Poultry. Ethology. 2001;107: 753–765.

58. Gunnarsson S, Matthews LR, Foster TM, Temple W. The demand for straw and feathers as litter substrates by laying hens. Appl Anim Behav Sci. 2000;65: 321–330.

59. Matthews LR, Chandler JH. Assessing behavioural demand functions: issues. In: Duncan IJH, Widowski TM, Haley DB, editors. Proceedings of the 30th International Congress of the International Society for Applied Ethology. Campbell Centre for the Study of Animal Welfare; 1996. p. 54.

60. Dawkins MS, Beardsley T. Reinforcing properties of access to litter in hens. Appl Anim Behav Sci. 1986;15: 351–364.

61. Hemsworth PH, Edwards LE. Natural behaviours, their drivers and their implications for laying hen welfare. Anim Produc Sci. 2020 [cited 18 Sep 2020]. doi:10.1071/AN19630

62. Campbell DLM, Makagon MM, Swanson JC, Siegford JM. Laying hen movement in a commercial aviary: Enclosure to floor and back again. Poult Sci. 2016;95: 176–187.

63. Campbell DLM, Ali ABA, Karcher DM, Siegford JM. Laying hens in aviaries with different litter substrates: Behavior across the flock cycle and feather lipid content. Poult Sci. 2017;96: 3824–3835.

64. Nicol CJ, Caplen G, Statham P, Browne WJ. Decisions about foraging and

risk trade-offs in chickens are associated with individual somatic response profiles. Anim Behav. 2011;82: 255–262.

65. Nicol C, Caplen G, Edgar J, Richards G, Browne W. Relationships between multiple welfare indicators measured in individual chickens across different time periods and environments. Anim Welf. 2011;20: 133–143.

66. Nicol CJ, Guilford T. Exploratory activity as a measure of motivation in deprived hens. Anim Behav. 1991;41: 333–341.

67. Lindberg AC, Nicol CJ. Space and density effects on group size preferences in laying hens. Br Poult Sci. 1996;37: 709–721.

68. Dawkins M. Priorities in the cage size and flooring preferences of domestic hens. Br Poult Sci. 1981;22: 255–263.

69. Grandin T, Johnson C. Animals Make Us Human: Creating the Best Life for Animals (Chapter 7: Chickens and other Poultry). First Edition. Mariner Books; 2010.

70. de Jong IC, Wolthuis-Fillerup M, van Reenen CG. Strength of preference for dustbathing and foraging substrates in laying hens. Appl Anim Behav Sci. 2007;104: 24–36.

71. Rodenburg TB, Tuyttens FAM, de Reu K, Herman L, Zoons J, Sonck B. Welfare assessment of laying hens in furnished cages and non-cage systems: an on-farm comparison. Anim Welf. 2008;17: 363–373.

72. Ali ABA, Campbell DLM, Karcher DM, Siegford JM. Influence of genetic strain and access to litter on spatial distribution of 4 strains of laying hens in an aviary system. Poult Sci. 2016;95: 2489–2502.

73. Schuck-Paim C, Alonso WJ. Egg Peritonitis Syndrome: A Painful and Prevalent Disease in Commercial Layers. In: Schuck-Paim C, Alonso WJ, editors. Quantifying Pain in Laying Hens. Independently published; 2021.

74. Daigle C, Siegford J. Welfare Quality® parameters do not always reflect hen behaviour across the lay cycle in non-cage laying hens. Anim Welf. 2014.

75. van Liere DW. The Significance of Fowls' Bathing in Dust. Anim Welf. 1992;1: 187–202.

76. Olsson IAS, Keeling LJ. Why in earth? Dustbathing behaviour in jungle and domestic fowl reviewed from a Tinbergian and animal welfare perspective. Appl Anim Behav Sci. 2005;93: 259–282.

77. Martin CD, Mullens BA. Housing and dustbathing effects on northern fowl mites (Ornithonyssus sylviarum) and chicken body lice (Menacanthus stramineus) on hens. Med Vet Entomol. 2012;26: 323–333.

78. Vezzoli G, Mullens BA, Mench JA. Dustbathing behavior: Do ectoparasites matter? Appl Anim Behav Sci. 2015;169: 93–99.

79. Carol Petherick J, Seawright E, Waddington D, Duncan IJH, Murphy LB. The role of perception in the causation of dustbathing behaviour in domestic fowl. Anim Behav. 1995;49: 1521–1530.

80. Vestergaard K. Dust-bathing in the domestic fowl — diurnal rhythm and dust deprivation. Applied Animal Ethology. 1982;8: 487–495.

81. Widowski TM, Duncan IJ. Working for a dustbath: are hens increasing pleasure rather than reducing suffering? Appl Anim Behav Sci. 2000;68: 39–53.

82. Wichman A, Keelin LJ. The influence of losing or gaining access to peat on the dustbathing behaviour of laying hens. Anim Welf. 2009;18: 149–157.

83. Olsson IAS, Keeling LJ, Duncan IJH. Why do hens sham dustbathe when they have litter? Appl Anim Behav Sci. 2002;76: 53–64.

84. Colson S, Arnould C, Michel V. Motivation to dust-bathe of laying hens housed in cages and in aviaries. Animal. 2007;1: 433–437.

85. Zimmerman PH, Koene P, van Hooff JA. Thwarting of behaviour in different contexts and the gakel-call in the laying hen. Appl Anim Behav Sci. 2000;69: 255–264.

86. Vestergaard KS, Kruijt JP, Hogan JA. Feather pecking and chronic fear in groups of red junglefowl: their relations to dustbathing, rearing environment and social status. Anim Behav. 1993;45: 1127–1140.

87. Faure JM. Rearing conditions and needs for space and litter in laying hens. Appl Anim Behav Sci. 1991;31: 111–117.

88. Lagadic H, Faure J-M. Preferences of domestic hens for cage size and floor types as measured by operant conditioning. Appl Anim Behav Sci. 1987;19: 147–155.

89. de Jong IC, Wolthuis-Fillerup M, van Reenen CG. Laywel: welfare implications of changes in production systems for laying hens. Deliverable 4.3 Report on substrate needs and preferences. 2007.

90. Orság J, Brouček J, Mačuhová L, Knížatová M, Flak P, Hanus A. Behaviour

of hens deprived of dustbathing. 2011 [cited 24 Sep 2020].

91. Keeling LJ. Inter-bird distances and behavioural priorities in laying hens: the effect of spatial restriction. Appl Anim Behav Sci. 1994;39: 131–140.

92. Vestergaard KS, Damm BI, Abbott UK, BildsøE M. Regulation of dustbathing in feathered and featherless domestic chicks: the Lorenzian model revisited. Anim Behav. 1999;58: 1017–1025.

93. Tauson R. Management and housing systems for layers – effects on welfare and production. null. 2005;61: 477–490.

94. Cooper JJ, Albentosa MJ, Redgate SE. The 24 hour activity budgets of hens in furnished cages. Br Poult Sci. 2004;45 Suppl 1: S38–40.

95. Scholz B, Kjaer JB, Petow S, Schrader L. Dustbathing in food particles does not remove feather lipids. Poult Sci. 2014;93: 1877–1882.

96. Campbell DLM, Makagon MM, Swanson JC, Siegford JM. Litter use by laying hens in a commercial aviary: dust bathing and piling. Poult Sci. 2016;95: 164–175.

97. Alvino GM, Tucker CB, Archer GS, Mench JA. Astroturf as a dustbathing substrate for laying hens. Appl Anim Behav Sci. 2013;146: 88–95.

98. Duncan IJ. Thirty years of progress in animal welfare science. J Appl Anim Welf Sci. 1998;1: 151–154.

99. Dawkins MS, Hardie S. Space needs of laying hens. Br Poult Sci. 1989;30: 413–416.

100. Appleby MC, Hughes BO. Welfare of laying hens in cages and alternative systems: environmental, physical and behavioural aspects. Worlds Poult Sci J. 1991;47: 109–128.